

D4 w1 Strict Bound -- POSITIVE Machine Certificate (2026-06-13)

Opera Numerorum | David J. Fox | June 13, 2026

STATUS: D4_THRESHOLD_PROVED -- Lean 4, classical trio, 0 sorry, 0 axiom

PROVED: $w1_weyl_series(11/5) < 1/7$

D4 threshold established for $\beta \geq 11/5 = 2.20$.

Supersedes D4_w1_NEGATIVE_Certificate (2026-06-01): negative at $\beta=0.86$.

D4 LEAN FILE SHA-256 (Towers/YM/W1_One_Seventh_Proof.lean)
44e7e2c6f3201c0f79e173245aba13dfcd281bdc

1. Statement Proved

theorem w1_seven_lt : $w1_weyl_series(11/5 : \mathbb{R}) < 1/7$

Proved in: Towers/YM/W1_One_Seventh_Proof.lean

Namespace: TheoremaAureum.Towers.YM.W1SeventhProof

Axiom footprint: {propext, Classical.choice, Quot.sound} -- classical trio only.

Sorry count: 0. New axioms: 0. Gaps: 0.

The theorem proves that the SU(3) Weyl-Toeplitz Bessel series at $\beta = 11/5$ (exact rational, $= 2.20$) is strictly less than $1/7$. This establishes the D4 threshold in the Kotecky-Preiss chain: for any $\beta \geq 11/5$ the polymer activity $w1(\beta) < 1/7 < 1/7$ guarantees that the Brydges-Federbush D2 rate condition $I > \log 7$ holds.

2. Proof Method

Same Taylor-Lagrange $N=28$ + Bessel interval enclosure pattern as KP_W1_Proof.lean (which proved $w1(30029/6250) < 1/56$ at the KP threshold).

Key constants at $\beta_7 = 11/5$:

β_7_rat	$= 11/5$	(exact rational)
r_7	$= 11/30 < 1/2$	(Bessel ratio)
q_7	$< 1/8$	(tail geometric ratio)
C_exp_7	$< 3/2$	(exp upper bound)
window S6k7	$= k \text{ in } \{-5..5\}$	(11 Bessel terms, $k \text{ in } [-5,5]$)
$tail_7_ub$	$= 3/200$	(outer Bessel tail bound)
N_bessel	$= 40$	(Taylor order for exp enclosure)

The decide-compatible rational value for seven_part_c has denominator $\sim 10^818$ digits (comparable to the KP case at $\sim 10^460$). Margin: LHS approx 0.1304, $1/7$ approx 0.1429, margin approx 0.0124.

3. D Series Chain Status

Ref	Name	Status	Lean File SHA (first 12)
D1	peierls_branching_bound	PROVED	b651950b71a4
D2	geometric_activity_bound	PROVED	b651950b71a4
D3	kp_summable (abstract)	PROVED	b651950b71a4
D4	$w1_weyl_series(11/5) < 1/7$	PROVED	44e7e2c6f320
D5	TruncatedActivityBound (concrete)	OPEN	--

D6	KP-summability -> spectral decay	OPEN	--
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4. SHA Chain

Item	SHA-256 (full)
D4 Lean file (W1_One_Seventh_Proof.lean)	44e7e2c6f3201c0f79e173245aba13dfcd281bdc
D1-D3 Lean file (BrydgesFederbush_D1D3.lean)	b651950b71a4c7b92726c681fa8ed5b93cf24418
KP_Closure.lean (W1_one_seventh_Surface CLOSED)	b13bf336cc7173b1df9a87174b1574b681a1f8a6
D4 NEGATIVE cert (superseded, 2026-06-01)	9a794ccf0c707812e6fa3db2095a350f2d5b61a011fcc77e453c548716ac8764

5. What This Does NOT Claim

NO mass gap. NO spectral gap. NO Clay result. NO Surface #1 discharge.

w1_weyl_series is the SU(3) Bessel series representation; not connected to the abstract N or hOS/h_bridge in Surface #1.

D5 (TruncatedActivityBound concrete instantiation) and D6 (KP-summability implies spectral decay) remain OPEN.

kotecky_preiss_criterion_Surface (Transfer.lean:409) stays OPEN.

Clay Millennium Problem (YM mass gap): OPEN throughout.

Even with D1-D4 all proved, the chain D4 -> D5 -> D3 -> D6 -> kotecky_preiss_criterion_Surface still requires D5 and D6 (Osterwalder-Seiler Ursell/cluster step and Brydges-Federbush decay bridge), both OUT-OF-TOWER and absent from Mathlib v4.12.0.

6. Honesty Locks

status_of_this_cert: D4_THRESHOLD_PROVED

supersedes: D4_w1_NEGATIVE_Certificate (2026-06-01, SHA 9a794ccf...)

closes_kotecky_preiss_criterion_Surface: false

closes_mass_gap: false

clay_ym_status: OPEN

axiom_footprint: {propext, Classical.choice, Quot.sound}

sorry_count: 0

new_axiom_count: 0

in_tower: Towers/YM/KP_Closure.lean -- W1_one_seventh_Surface CLOSED

repo: <https://github.com/DavidFox998/Yang-Mills-MassGap>

Series: Opera Numerorum -- Battle Plan v1.6

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